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Neutron detection element

Abstract

A neutron detection element **100** includes: a neutron detector configured to detect a neutron and convert the neutron into an electrical signal; and an amplifier configured to amplify an output of the neutron detector. The neutron detector includes: a semiconductor layer of a first conductivity type; a detector diffusion layer of a second conductivity type in the semiconductor layer; and a neutron conversion layer on the detector diffusion layer. The neutron conversion layer converts a neutron into an α-ray. The amplifier includes a plurality of transistors in the semiconductor layer. The neutron conversion layer is a metal film including a layer containing boron 10 or a layer containing lithium 6.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2011/0127527	12/2010	Kub	257/E31.086	G01T 3/08
2014/0299781	12/2013	Hultman et al.	N/A	N/A
2016/0172396	12/2015	Masuda	257/295	H10F 30/292
2017/0260619	12/2016	Störmer et al.	N/A	N/A
2017/0363755	12/2016	Ottaviani et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H07-176777	12/1994	JP	N/A
2012-181065	12/2011	JP	N/A
2016-535240	12/2015	JP	N/A
2018-017613	12/2017	JP	N/A
2018-505396	12/2017	JP	N/A
2018-141749	12/2017	JP	N/A
2013/002697	12/2012	WO	N/A

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a neutron detection element and a two-dimensional neutron sensor.

BACKGROUND ART

(2) Neutrons are particles without charges and have the characteristics of passing through various

substances. Neutrons are thus used in neutron radiography for observing the inside of a substance. In the use of neutrons for internal observation or treatment, detecting transmitted neutrons is important.
(3) There is also a study using neutrons for cancer treatment, such as boron neutron capture therapy (BNCT). Even in such a case, detecting how the projected neutrons are absorbed is important.
(4) Known techniques for neutron detection are gas discharge, scintillation, or imaging plates, for example. In recent years, a neutron detection element made of semiconductor has also been studied (see, e.g., Patent Document 1).
CITATION LIST
Patent Document
(5) Patent Document 1: Japanese Unexamined Publication No. 2012-181065
SUMMARY OF THE INVENTION
Technical Problems
(6) A typical neutron detection element made of semiconductor generates charged particles from neutrons and detects the generated charged particles. In generating the charged particles from neutrons, the neutrons are incident on boron 10 (.sup.10B), for example, to generate helium nuclei (i.e., α-rays) and lithium nuclei (i.e., Li-particle beams). The α-rays and Li-particle beams incident on a depletion layer causes generation of electron-hole pairs. Thus, by measuring a current generated by the caused electron-hole pairs, the neutrons can be detected. When neutrons are incident on .sup.10B, however, the α-rays and the Li-particle beams are emitted in opposite directions and thus incident on the depletion layer. If incident, the α-rays and the Li-particle beams cause different amounts of charges in the depletion layer. The output of the detector thus varies depending on the incident charged particle beams.
(7) It is an objective of the present disclosure to achieve a neutron detection element made of semiconductor causing less variation in output.
Solution to the Problems
(8) A neutron detection element according to an aspect of the present disclosure includes: a neutron detector configured to detect a neutron and convert the neutron into an electrical signal; and an amplifier configured to amplify an output of the neutron detector. The neutron detector includes: a semiconductor layer of a first conductivity type; a detector diffusion layer of a second conductivity type in the semiconductor layer; and a neutron conversion layer on the detector diffusion layer. The neutron conversion layer converts a neutron into an α-ray. The amplifier includes a plurality of transistors in the semiconductor layer. The neutron conversion layer is a metal film including a layer containing boron 10 or a layer containing lithium 6.
Advantages of the Invention
(9) The neutron detection element according to the present disclosure allows the incidence of only α-rays on a semiconductor layer and causes less output variation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a circuit diagram showing a neutron detection element according to an embodiment.
(2) FIG. **2** is a cross-sectional view showing the neutron detection element according to the embodiment.
(3) FIG. **3**A is a cross-sectional view showing a manufacturing process of the neutron detection element according to the embodiment.
(4) FIG. **3**B is another cross-sectional view showing the manufacturing process of the neutron detection element according to the embodiment.
(5) FIG. **3**C is still another cross-sectional view showing the manufacturing process of the neutron

detection element according to the embodiment.
(6) FIG. **3**D is yet another cross-sectional view showing the manufacturing process of the neutron detection element according to the embodiment.
(7) FIG. **3**E is further another cross-sectional view showing the manufacturing process of the neutron detection element according to the embodiment.
(8) FIG. **4** is a plan view showing a two-dimensional neutron sensor according to the embodiment.
(9) FIG. **5** is a cross-sectional view showing a neutron detection element according to a variation.
DESCRIPTION OF EMBODIMENTS
(10) As shown in FIG. **1**, a neutron detection element **100** according to an embodiment includes a neutron detector **101** configured to detect neutrons and convert the neutrons into electrical signals, and an amplifier **102** configured to amplify the outputs of the neutron detector. The amplifier **102** includes a reset transistor (RST), a source follower transistor (SF), and a row selection transistor (RS). The neutron detection element can be driven by applying a reset signal to φR and a row selection signal to φX.
(11) As shown in FIG. **2**, the neutron detector **101** and the amplifier **102** are formed in a p.sup.−-type semiconductor layer **112** on an n-type semiconductor substrate **111**. In this embodiment, the semiconductor substrate **111** and the semiconductor layer **112** are made of silicon carbide (SiC).
(12) The semiconductor layer **112** includes an n.sup.+-type detector diffusion layer **113**, source/drain diffusion layers **121**, and a p.sup.+-type potential stabilizing region **114**. On the semiconductor layer **112**, a gate insulating film **123** is formed. In a predetermined region on the gate insulating film **123**, a gate electrode **124** is formed. An ohmic electrode **117** in ohmic contact with the potential stabilizing region **114**, and source/drain electrodes **122** in ohmic contact with the source/drain diffusion layers **121** are also formed. The ohmic electrode **117** and the source/drain electrodes **122** include a silicide layer **141** in ohmic contact with the diffusion layers and a metal electrode layer **142** covering the silicide layer **141**.
(13) On the semiconductor layer **112**, an interlayer insulating film **133** is formed to cover the electrodes. On the interlayer insulating film **133**, a neutron conversion layer **115** and interconnects **135** and **136** are formed. Accordingly, the neutron detector **101** and the amplifier **102** are formed such that the amplifier **102** includes the reset transistor (RST), the source follower transistor (SF), and the row selection transistor (RS). In this embodiment, one of the source/drain diffusion layers of the reset transistor (RST) is formed integrally with the detector diffusion layer **113**.
(14) On the back surface of the semiconductor substrate **111**, a back surface electrode **126** is provided. The back surface electrode **126** is an ohmic electrode in ohmic contact with the back surface of the semiconductor substrate **111** and may be made of niobium-nickel silicide, for example. The back surface electrode **126** is advantageous in quickly discharging the electron and hole carriers generated in the semiconductor substrate **111** and stabilizing device operation.
(15) The neutron conversion layer **115** is a metal film including, as a conversion functional layer, a .sup.10B-containing layer **115**A containing boron 10 (or .sup.10B) which is an isotope of boron. The .sup.10B-containing layer **115**A may be formed, for example, by ion implantation of .sup.10B or deposition of .sup.10B through sputtering or vapor deposition, for example. Instead of the .sup.10B-containing layer, the conversion functional layer may be a layer containing lithium 6 (.sup.6Li) which is an isotope of lithium.
(16) While there is a lower metal layer **115**B under the .sup.10B-containing layer **115**A and an upper metal layer **115**C on the .sup.10B-containing layer **115**A in this embodiment, there may be no upper metal layer **115**C. The lower metal layer **115**B and the upper metal layer **115**C are preferably made of metal with a relatively large atomic number in order to allow less penetration of α-rays and Li-rays. For example, the metal can be aluminum (Al), tungsten (W), or molybdenum (Mo). Among these, Al can be formed in the same manner as an Al interconnect and is thus preferably employed.
(17) The neutrons incident on the neutron conversion layer **115** react with .sup.10B to generate α-

rays and Li-particle beams. In the lower metal layer **115**B made of Al, for example, the Li-particle beams attenuate more easily than the α-rays. Only the α-rays pass through the lower metal layer **115**B and are incident on the detector diffusion layer **113** to generate electron-hole pairs in the depletion layer near the PN junction between the n.sup.+-type detector diffusion layer **113** and the p-type semiconductor layer **112**. The negative charges (i.e., electrons) of the generated electron-hole pairs move to the n.sup.+-type detector diffusion layer **113**. These charges are accumulated in the gate electrode of the SF transistor through the interconnects **135** and **136** to change the potential of this gate electrode.
(18) In this embodiment, since there is the n-type semiconductor substrate **111** under the p-type semiconductor layer **112**, a parasitic bipolar junction transistor (BJT) is further formed. The positive charges (i.e., holes) from the electron-hole pairs generated in the depletion layer are injected into the base of the parasitic BJT which is then temporarily turned on. The electrons move from the semiconductor substrate **111** through the base to the detector diffusion layer **113**. Like the electrons of the electron-hole pairs generated in the depletion layer, these electrons are accumulated in the gate electrode of the SF transistor through the interconnects **135** and **136** to further change the potential of this gate electrode. At this time, a negative voltage is applied to the back surface electrode **126** of the semiconductor substrate **111**. The amplification effect of the parasitic BJT then further increases, that is, the number of electrons injected from the semiconductor substrate **111** into the detector diffusion layer **113** can increase. The parasitic BJT can increase the sensitivity of the sensor. The parasitic BJT is however not necessarily needed and this sensor operates even with the detector diffusion layer **113** only.
(19) The charges accumulated in the gate electrode of the SF transistor change the potential of this gate electrode, which results in turning the RS transistor on and then changing the output voltage in accordance with the potential of the gate electrode of the SF transistor. By measuring this potential, the number of neutrons incident on the sensor can be measured.
(20) In this embodiment, the detector diffusion layer **113** is integral with the source/drain diffusion layer of the reset transistor (RST). The negative charges moving to the detector diffusion layer **113** are read out by the amplifier **102** and a signal is obtained which corresponds to the energy of the neutrons incident on the neutron conversion layer **115**. On the other hand, the positive charges (i.e., holes) are discharged through the potential stabilizing region **114**. While the parasitic BJT increases the neutron detection sensitivity, a configuration may exclude the parasitic BJT.
(21) The neutron detection element **100** according to this embodiment detects the neutrons using only the α-rays out of the α-rays and the Li-particle beams generated by the neutrons incident on the neutron conversion layer **115**. This configuration allows improvement in the quantitativity of neutrons.
(22) In this embodiment, the neutron conversion layer **115** is formed only right above the detector diffusion layer **113**, and not above the transistors or other elements forming the amplifier **102**. The neutron conversion layer **115** only right above the detector diffusion layer **113** allows less α-rays emitted from the neutron conversion layer **115** to be incident on the regions of the semiconductor layer **112** other than the detector diffusion layer **113**. This configuration thus causes less malfunction. While the region “right above the detector diffusion layer **113**” overlaps the detector diffusion layer **113** in a plan view, the detector diffusion layer **113** and the neutron conversion layer **115** are not necessarily overlap each other completely. The neutron conversion layer **115** may be formed in a region other than the region right above the detector diffusion layer **113**.
(23) The silicon carbide light-receiving element according to this embodiment may be formed as follows, for example. First, as shown in FIG. **3**A, the p-type semiconductor layer **112** is epitaxially grown on the n-type semiconductor substrate **111**. Subsequently, n-type impurities are selectively implanted into the semiconductor layer **112** using a first ion implantation mask to form the n.sup.+-type detector diffusion layer **113** and the source/drain diffusion layers **121**. In addition, the p.sup.+-type potential stabilizing region **114** is formed using a second ion implantation mask. After that,

activation annealing is performed.
(24) While the material of the semiconductor substrate **111** is not particularly limited but may be 4H-SiC. The semiconductor layer **112** may have a thickness ranging from about 1 μm to about 5 μm, for example. The first and second ion implantation masks may be made of a silicon oxide (SiO.sub.2) film, for example. In the case of SiC semiconductor, the activation annealing may be heat treatment at a temperature of about 1700° C. after covering the semiconductor layer **112** subjected to ion implantation with a carbon cap film.
(25) Next, as shown in FIG. **3**B, thermal oxidation is performed to form the gate insulating film **123** with a thickness of about 20 nm and then form the ohmic electrode **117**, the source/drain electrodes **122**, the gate electrode **124**, and the back surface electrode **126**. The ohmic electrode **117** and the source/drain electrodes **122** may include the silicide layer **141** and the metal electrode layer **142**, for example. The silicide layer **141** may be formed as follows, for example. A metal film to serve as the silicide layer **141** is selectively formed by lift-off using a resist mask and then silicified by heat treatment. The metal film in this case may be a niobium-nickel film, for example, but may be a film, such as a nickel-molybdenum alloy film, made of another metal that forms silicide. The metal electrode layer **142** may be formed as follows. A metal film made of Al, for example, is formed to cover the silicide layer **141** and selectively removed by etching. The metal electrode layer **142** and the gate electrode **124** may be formed by the same process. The metal electrode layer **142** and the gate electrode **124** may be made of titanium nitride or polysilicon, for example. The back surface electrode **126** may be formed as follows. A metal film, such as a niobium-nickel film, is formed on the back surface of the semiconductor substrate **111** and then silicified.
(26) Next, as shown in FIG. **3**C, the interlayer insulating film **133** made of silicon oxide, for example, is formed on the entire surface of the semiconductor substrate **111**, and the lower metal layer **115**B is formed on the interlayer insulating film **133**. The thickness of the interlayer insulating film **133** influences the incidence of the α-rays emitted from the neutron conversion layer **115** on the detector diffusion layer **113** and may be about 1 μm, for example. After that, .sup.10B is ion-implanted into the lower metal layer **115**B to form the .sup.10B-containing layer **115**A which serves as an impurity diffusion layer. The lower metal layer **115**B may be any metal layer that attenuates the Li-particle beams and may be made of aluminum, tungsten, or molybdenum, for example. The .sup.10B-containing layer **115**A is not necessarily formed by ion implantation and may be formed by depositing .sup.10B thorough sputtering or vapor deposition, for example. The lower metal layer **115**B has a thickness allowing the transmission of no Li-particle beams but the α-rays. The thickness may be determined as appropriate in accordance with the type of the used metal. For example, in the case of aluminum, the thickness may range from about 2 μm to about 4 μm. In the case of tungsten, the thickness may be about 0.4 μm. When forming the .sup.10B-containing layer **115**A by ion implantation, the amount of implantation may range from about 1×10.sup.15 cm.sup.−2 to about 4×10.sup.15 cm.sup.−2. The thickness of the upper metal layer **115**C is not particularly limited but may range from about 10 nm to about 1000 nm in the case of aluminum, for example. The .sup.6Li-containing layer may be formed similarly.
(27) Next, as shown in FIG. **3**D, the lower metal layer **115**B with the .sup.10B-containing layer **115**A formed thereon is selectively removed except for the region right above the detector diffusion layer **113**. Next, as shown in FIG. **3**E, the upper metal layer **115**C and the interconnects **135** and **136** are formed. If made of the same metal, the upper metal layer **115**C and the interconnects **135** and **136** can be formed by the same process.
(28) Neutron detection elements **100**, each being the neutron detection element **100** according to this embodiment, may be arranged in a matrix as shown in FIG. **4** to form a two-dimensional neutron sensor **200** including the neutron detection elements **100** each serving as a pixel. With the use of the two-dimensional neutron sensor **200**, the distribution of neutrons on a plane can be easily measured and the transmission and absorption of neutrons can be evaluated in real time by neutron radiography or BNCT. In addition, with an excellent quantitativity, the neutron detection element

100 according to this embodiment allows accurate visualization of the intensity distribution of neutrons.

(29) As shown in FIG. **5**, a high-density layer **151** may be interposed between the neutron conversion layer **115** and the detector diffusion layer **113**. The high-density layer **151** can reduce the α-rays penetrating the detector diffusion layer **113**. This configuration allows further improvement in the detection sensitivity. The high-density layer **151** may be a layer having a higher density than the semiconductor layer **112**, and is preferably a layer containing metal. Examples include a simple metal layer, an alloy layer containing a plurality of metals, or a layer, such as a silicide layer, containing metal and non-metal. Among these, metal with an atomic number exceeding 14 is preferably contained. In view of the handleability, nickel, niobium, titanium, cobalt, or any other suitable metal is more preferably contained. These metals are preferably also employed in view of easier formation of silicide. If made of the same silicide layer as the source/drain electrodes **122**, the high-density layer can be formed without increasing the number of steps. In this case, the silicide layer **141** of the source/drain electrodes **122** and the high-density layer **151** can be formed integrally.

(30) The high-density layer **151** is preferably in contact with the detector diffusion layer **113**. Alternatively, another layer, such as an insulating film, may be interposed between the high-density layer **151** and the detector diffusion layer **113**. The high-density layer **151** is preferably formed at least right under the neutron conversion layer **115** in the detector diffusion layer **113**. The high-density layer **151** may cover the entire detector diffusion layer **113** or a region wider than the detector diffusion layer **113**. The thickness of the high-density layer **151** is not particularly limited but preferably ranges from about 0.01 μm to about 0.3 μm in order to allow the reach of the α-rays to the detector diffusion layer but no penetration. If formed by the same procedure as the silicide layer **141** of the source/drain electrodes **122**, the high-density layer **151** can have the same thickness as the silicide layer **141** of the source/drain electrodes **122**.

(31) In this embodiment, the semiconductor layer **112** is of the p-type and includes the transistors of the n-type. The semiconductor layer **112** is of the p-type and includes the p.sup.+-type potential stabilizing region **114**, which allows quick discharge of the holes caused by the α-rays and thus allows stabilization of the output potential of the detection element. Alternatively, the semiconductor layer **112** may be of the n-type and include the detector diffusion layer and the transistors of the p-type. In this case, the potential stabilizing region may be of the n.sup.+-type. Note that the potential stabilizing region may or may not be provided as necessary. In this embodiment, the potential stabilizing region **114** is spaced apart from and adjacent to the detector diffusion layer **113**. Such a configuration allows an increase in the advantage of stabilizing the potential. The potential stabilizing region **114** can be formed in any part of the semiconductor layer **112** which is the same as the detector diffusion layer **113**. The sign “+” following “p” or “n” indicating the conductivity type represents a higher impurity concentration than the case without any sign, while the sign “−” represents a lower impurity concentration than the case without any sign.

(32) As in this embodiment, the radiation resistance can be improved by using SiC semiconductor, which can largely extend the lifetime of the element. Alternatively, the element may be made of silicon semiconductor.

INDUSTRIAL APPLICABILITY

(33) The neutron detection element according to the present disclosure causes less variation in the output and facilitates quantitative detection of neutrons, and is thus useful particularly in the medical or other field using neutrons.

DESCRIPTION OF REFERENCE CHARACTERS

(34) **100** Neutron Detection Element **101** Neutron Detector **102** Amplifier **111** Semiconductor Substrate **112** Semiconductor Layer **113** Detector Diffusion Layer **114** Potential Stabilizing Region **115** Neutron Conversion Layer **115**A .sup.10B-Containing Layer **115**B Lower Metal Layer **115**C

Upper Metal Layer **117** Ohmic Electrode **121** Source/Drain Diffusion Layer **122** Source/Drain Electrode **123** Gate Insulating Film **124** Gate Electrode **126** Back Surface Electrode **133** Interlayer Insulating Film **135** Interconnect **136** Interconnect **141** Silicide Layer **142** Metal Electrode Layer **151** High-Density Layer **200** Two-Dimensional Neutron Sensor

Claims

1. A neutron detection element comprising: a neutron detector configured to detect a neutron and convert the neutron into an electrical signal; and an amplifier configured to amplify an output of the neutron detector, the neutron detector including: a semiconductor layer of a first conductivity type; a detector diffusion layer of a second conductivity type in the semiconductor layer; a neutron conversion layer on the detector diffusion layer, the neutron conversion layer being configured to convert a neutron into an x-ray; and a high-density layer between the neutron conversion layer and the detector diffusion layer, the high-density layer being made of a material having a higher density than the semiconductor layer, the amplifier including a plurality of transistors in the semiconductor layer, the neutron conversion layer being a metal film including a conversion functional layer containing boron 10 or a layer containing lithium 6.
2. The neutron detection element of claim 1, wherein the layer containing boron 10 or the layer containing lithium 6 is an impurity diffusion layer.
3. The neutron detection element of claim 1, wherein the semiconductor layer is a silicon carbide semiconductor layer.
4. The neutron detection element of claim 1, wherein the neutron conversion layer is formed right above the detector diffusion layer with an insulating film interposed therebetween.
5. The neutron detection element of claim 1, wherein the high-density layer is a metal silicide layer.
6. The neutron detection element of claim 1, wherein the neutron detector includes a potential stabilizing region of the first conductivity type in the semiconductor layer, the potential stabilizing region having a higher impurity concentration than the semiconductor layer.
7. A two-dimensional neutron sensor comprising: neutron detection elements, each being the neutron detection element of claim 1, in a matrix.
8. A neutron detection element comprising: a neutron detector configured to detect a neutron and convert the neutron into an electrical signal; and an amplifier configured to amplify an output of the neutron detector, the neutron detector including: a semiconductor layer of a first conductivity type; a detector diffusion layer of a second conductivity type in the semiconductor layer; a neutron conversion layer on the detector diffusion layer, the neutron conversion layer being configured to convert a neutron into an x-ray; and a potential stabilizing region of the first conductivity type in the semiconductor layer, the potential stabilizing region having a higher impurity concentration than the semiconductor layer, the amplifier including a plurality of transistors in the semiconductor layer, the neutron conversion layer being a metal film including a conversion functional layer containing boron 10 or a layer containing lithium 6, wherein the semiconductor layer is a silicon carbide semiconductor layer.
9. The neutron detection element of claim 8, wherein the semiconductor layer is provided on a semiconductor substrate that has no buried insulator.
10. The neutron detection element of claim 1, wherein the plurality of transistors includes transistor diffusion layers, and wherein both of the detector diffusion layer and the transistor diffusion layers are formed in the semiconductor layer provided on a first side of a substrate.
11. The neutron detection element of claim 1, wherein the plurality of transistors includes transistor diffusion layers, and wherein the detector diffusion layer is integral with one of the transistor diffusion layers.
12. The neutron detection element of claim 1, wherein the conversion functional layer is sandwiched between lower metal layer and upper metal layer.

13. The neutron detection element of claim 1, wherein the high-density layer is a layer containing metal.

14. The neutron detection element of claim 1, wherein the high-density layer is a silicide layer.

15. The neutron detection element of claim 1, wherein the semiconductor layer is provided on a semiconductor substrate that has no buried insulator.